

Chaitanya Kumar Battula

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SUMMARY

Data Analyst with a strong foundation in statistical programming, data wrangling, and clinical data analysis. Proficient in SAS and Python, with experience building predictive models, performing statistical validation, and working with healthcare datasets. Skilled in writing clean, validated code for data analysis and visualization, and passionate about contributing to biostatistics and clinical research. Proficient in SAS for data manipulation, validation, and report generation; familiar with PROC SQL, DATA steps, and statistical procedures used in clinical datasets.

EDUCATION

Master in Computer Science, Purdue University (GPA: 3.80)	United States
Post Graduation Program in Artificial Learning and Machine Learning, Purdue University	United States
Bachelor in Computer Science, JSS Science & Technology University(GPA: 3.70)	India

WORK EXPERIENCE

Machine Learning Engineer - Internship	Jun '24 — Feb '25
Reality AI Lab	New York, United States

- Developed REX, an AI tutor, leveraging Generative AI, RAG, and embeddings, improving user engagement by 40% through personalized worksheet recommendations.
- Designed an AI-powered worksheet generation platform, optimizing content creation efficiency by 60%, streamlining customization, distribution, and accessibility for educational content.
- Implemented LangChain with OpenAI embeddings, reducing content retrieval latency by 30%, ensuring context-aware and high-quality worksheet creation. Built data-driven analytics pipelines on Google Cloud (GCP), accelerating insights generation by 50%, enabling automated decision-making and dynamic worksheet adaptation.

Machine Learning Engineer - Internship	Sep '20 — Oct '20
Finland Labs	India

- Developed predictive models and statistical algorithms, improving forecasting accuracy by 18% and reducing data processing time by 35%. Designed and implemented data integration pipelines on AWS, ensuring seamless extraction, transformation, and storage for analytical workflows.
- Conducted A/B testing and statistical analysis on large datasets, analyzing 423,761 events to optimize decision-making strategies. Processed and analyzed structured and unstructured data using AWS cloud-based tools, ensuring high-quality datasets for machine learning applications and business insights.

PROJECT EXPERIENCE

Health Insurance Claim Prediction (*Classification, Risk Modeling, Healthcare Data Insights*) [Link](#)

- Built a machine learning model to predict the likelihood of insurance claims, improving fraud detection and risk assessment processes. Preprocessed and cleaned a dataset of 10,000+ records, addressing missing values and engineering features to enhance accuracy. Handled class imbalance with SMOTE, improving recall by 15% compared to the baseline model. Developed and evaluated models including Logistic Regression, Random Forest, and Gradient Boosting, achieving 88% accuracy, 85% precision, and an ROC-AUC of 0.90 with Gradient Boosting.

Medical Cost Analysis for Smokers and Non-Smokers [Link](#)

- Analyzed medical costs for over 1,300 patients to identify key cost drivers, such as smoking status, BMI, and age.
- Conducted exploratory data analysis using Seaborn and Matplotlib to uncover trends and compare costs between smokers and non-smokers. Built and evaluated regression models, achieving an R-squared of 0.75 for predicting medical costs. Provided actionable insights to optimize healthcare spending and resource allocation.

Breast Cancer Prediction (*Predictive Modeling, Medical Classification*) [Link](#)

- Built a machine learning model to predict breast cancer likelihood, achieving 94% accuracy, 92% recall, and an ROC-AUC of 0.93 with SVM. Preprocessed samples, addressing missing values, scaling features, and improving accuracy by 12% through feature engineering and hyperparameter tuning.
- Evaluated model performance using Confusion Matrix, Precision, Recall, and F1-score to ensure robust validation.

Heart Attack Analysis(*Clinical Data, Cardiology, Risk Prediction, Feature Importance, Bioinformatics,EDA*)[Link](#)

- Developed a machine learning model to predict the risk of heart attacks based on patient data, supporting early diagnosis and treatment planning. Analyzed a dataset patient records, identifying key risk factors such as cholesterol levels, age, and blood pressure through exploratory data analysis. Built and evaluated Logistic Regression, Random Forest, and SVM models, with Random Forest achieving 90% accuracy, 88% precision, and an ROC-AUC of 0.92. Improved model interpretability by performing feature importance analysis, reducing misclassification rates by 10%.

Customer Segmentation (Clustering, Unsupervised Learning) [Link](#)

- Engineered a comprehensive suite of advanced clustering algorithms for segmenting bank customers based on behavior, demographics, and transaction history. Implemented K-means, Hierarchical, DBSCAN, Gaussian Mixture Models, Mean Shift, Agglomerative, and Affinity Propagation, optimizing each for specific data characteristics. Achieved a best Silhouette Score of 0.42 and a Davies-Bouldin Index of 0.83, indicating high cluster quality and separation.

Customer Life Value Prediction (Regression, Ensemble Learning, SHAP/LIME) [Link](#)

- Developed an advanced suite of predictive models using Jupyter Notebook for Customer Lifetime Value (CLV) using sophisticated ensemble techniques, including Gradient Boosting Machines (GBM) and Random Forests. Conducted extensive feature engineering, dealt outliers with Isolation forest, obtained best features based on feature importance, hyper tuned and evaluated model performance with regression metrics such as Mean Absolute Error (MAE) of 4 and an R-squared (R²) score of 0.99, demonstrating high precision and predictive accuracy.
- Obtained LIME and SHAP values to explain the advanced interpretability of the model.

A/B Testing Recommender System: Conversion Rates & User Behavior: (Hypothesis testing, Statistical Testing)

- Conducted A/B test on a recommender system, analyzing 423,761 events from 2,594 users to evaluate conversion rates.
- Applied statistical tests (Z-tests, Mann-Whitney U), revealing a 20% increase in product page conversion for Group B, with stable cart and purchase conversion rates. Used funnel & time-series analysis to visualize early-stage engagement trends and assess impact on user behavior.

Supermarket Sales & Customer Insights Visualization (Tableau, KPIs, Cost Trends, Dashboards) [Link](#)

- Designed interactive dashboards in Tableau for real-time analysis of sales performance and customer trends. Developed KPIs for sales, orders, and profit, integrating multiple data sources to enable interactive drill-downs and advanced filtering. Conducted customer segmentation and sales trend analysis,optimizing marketing strategies and product planning.

CERTIFICATION:

SAS Programming BASE Certification – <i>Udemy</i>	Mar 2024
Data Analyst with R – <i>DataCamp</i>	Dec 2022
AI/ML – Advanced ML and Signal Processing – <i>Coursera</i>	Aug 2023
Power BI / Tableau – <i>Udemy</i>	Mar 2024

SKILLS

Programming Languages: Python, Julia, R, C, C++
Data Visualization: Power BI, SAS, Microsoft Excel (VLOOKUPs, conditional formatting), Tableau, Dashboards
ML Frameworks: TensorFlow, Keras, scikit-learn, Pandas, NumPy, Matplotlib,Seaborn, Plotly OpenCV, NLTK
Forecasting Methods: Moving Averages, Exponential Smoothing, ARIMA, SARIMA, Prophet
Advanced Analytics: Machine Learning (ML), Deep Neural Networks (DNN), CNN, Advanced Computer Vision & Signal Processing, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Optimization Algorithms,, Statistical Methods, Regression Models, Quantitative Models, Time Series Analysis, Performance Metrics, Financial Data, ETL
Business Skills: Business Strategy, Financial Analysis, Sales Data Analysis, Productivity Trends, Revenue Forecasting
Cloud Technologies & Databases: AWS(S3, EC2, Sage Maker), GCP, Azure ,SQL, NoSQL, GIT